

Utsav Acharya

✉ clerisy47@gmail.com ☎ +9779860307243 📁 Portfolio 🌐 LinkedIn 🐙 Github

An ever learning **Computer Engineering** student with a strong foundation in **Mathematics**, **Data Science**, **Machine Learning**, and **Backend Development** fundamentals and person with an urge for continual learning, problem solving and innovation.

WORK EXPERIENCE

Student Trainee 04/2025 – 11/2025
NAAMI

Conducting computer vision research in the physics lab focused on:

- **Disease detection** in rice crops
- **Behavioral monitoring** of poultry

Backend Engineer and Data Scientist 11/2023 – 03/2025
Omdena 📁

Participated in the following real-world deployed projects:

- **Combatting illegal deforestation in Romania** 📁
- **Mapping Seagrass Meadows** 📁
- **Agriculture Analysis on Kenya** 📁

Data Science and Business Analytics Intern 09/2023 – 10/2023
The Sparks Foundation

- Created raspberry pi app for **Tomato Disease Classification** 📁
- Did **Exploratory Data Analysis on IPL 2023** 📁

ACADEMIC PROJECTS

Compliance Guide 📁

The Compliance Guide saves you countless hours by simplifying complex policy documents and compliance frameworks, helping you ensure seamless security and a gap-free product.

- Used **Llama** for summariser
- Increased accuracy to upto 95% using **RAG**
- Used locally hosted **Llama** as comparator
- Used **Streamlit** for creating dashboards

Volt Vision 📁

Energy demand forecasting for Nepal households.

- Used pandas for data analysis and **Xgboost** for demand forecasting.
- Used **Sarimax** and **Arimax** model for time series analysis and price forecasting.
- Deployment Pipelines for Energy Demand Forecasting System using ZenML/MLflow

Rakhsak 📁

An all-in-one safety application designed to provide **real-time updates** on accidents, floods, and other hazards.

- Used **Nextjs** for frontend
- Used **Fastapi** as backend with mongodb dataset
- Use **Beautiful Soup** for webscrapping

Guardian Shield 📁

Guardian-Shield is a browser extension designed to enhance your online security by detecting and preventing common fraud and phishing attempts in your emails.

- Used **Chrome apis** for extension
- Used **Bert Transformer** and **Pytorch** for spam detection
- Use **Peagsus Transformer** and pytorch for terms and conditions summarizing.

EDUCATION

Bachelor's in Computer Engineering 2023 – present
Pulchowk Engineering Campus Kathmandu, Nepal
Average Percentage: 82.4

High School 2021 – 2023
Capital College and Research Centre Kathmandu, Nepal
GPA: 3.91 + 3.73

SKILLS

Languages — Python, C, C++, JavaScript

Frameworks — Numpy, Pandas, Flask, Fastapi, Django, Matplotlib, Seaborn, Scikit Learn, TensorFlow, OpenCV, spaCy, React

Developer Tools — Git, Docker, VS Code, Tableau

AWARDS

National IT Olympiad Silver Medalist <i>STEM, Nepal</i>	2022
National Hackathon Winner <i>LOCUS</i>	2024
Runner UP:- Nepal Hacks <i>Security Pal, NAAMI</i>	2024
Specialist, National top 10 <i>Codeforces</i>	2024
Tribhuvan University Merit-Based Scholarship <i>Tribhuvan University</i>	2023
National Physics-Mathematics Olympiad Bronze Medalist <i>SOMES Nepal</i>	2021

CERTIFICATES

- | | | |
|---|---|--|
| • Google Advanced Data Analytics 🔗 | • IBM Data Science Professional Certificate 🔗 | • Stanford Machine Learning Specialization 🔗 |
| • Tensorflow Developer Professional Certificate 🔗 | • Meta Backend Professional Certificate 🔗 | |

RESEARCH INTERESTS

- | | | |
|--|--|---|
| • Computer Vision for Agriculture and Environmental Monitoring | • Natural Language Processing for Compliance and Policy Analysis | • AI-based Risk Assessment and Automation |
|--|--|---|

LEADERSHIP & EXTRACURRICULARS

Data Research Council For Students <i>Core Member</i> • Mentored 200+ students in intermediate Python, providing guidance and support to help them improve their skills. • Managed GitHub repositories, social media accounts, and Discord servers, ensuring that these platforms were up-to-date and effectively engaged with our community.	11/2022 – 12/2023
Together We Learn <i>Co founder</i> • Co-founded and led an organization of 500+ members • Conducted 5+ webinars on programming-related topics, sharing knowledge and insights with a broad audience. • Wrote 10+ newsletters focused on data science and programming in general, providing valuable information and updates to 200+ subscribers.	03/2022 – 08/2023